Flight Dynamics of the X-57 Maxwell in JSBSim

A built-up model with fourteen propellers, proved on the Tecnam P2006T and checked against published CFD, tests and simulation

Technical report • November 2026

Abstract

A flight-dynamics model of the X-57 Maxwell in its Mod III and Mod IV forms has been written in JSBSim for the FlightGear simulator. Mods III and IV never flew, so nothing here can be compared with a flight of those aircraft: the references are the CFD database, wind tunnel and vibration tests of their parts, and piloted-simulation results, all published. The model is built up part by part, 17 wing strips a side, 3 stabilator strips a side, 4 fin strips and 11 fuselage stations, so that any one of the fourteen propellers can fail and leave its own part of the wing. Each wing strip sits in the slipstream of its own disc, carried by the ring-vortex wake that takes a rotor's downwash to the airframe in the same author's helicopter models, with one fitted constant, in place of the three a plain jet needed. The method was first proved on the Tecnam P2006T, whose fuselage and tail the X-57 keeps: it passes that aircraft's flight manual, cruise within 2.7 kt at 27 points and climb on one and two engines within limits. Its flight-test misses are traced to the test paper's methods by a second source: at the Mod II simulator's mass the model gives that simulator's short period within 5 %, and the paper's own derivatives give 4.95 rad/s where its flight figure is 3.41. On the X-57 the model reproduces the CFD's clean and flapped lift, the propellers' added lift inside the solvers' own spread, a cruise propeller's lift, drag reduction, pitch and roll on its wing within 21 % unfitted, and cruise L/D of 14.2 against 13.5. Its modes follow the published simulator's within 0.04 of damping and 10 % of frequency for Mod III. A motor lost with no pilot follows the published time histories: at 3 s bank −74°, pitch −9.9°, heading −23.2° against about −87°, −11°, −23°, once the run is flown in the configuration the published traces show rather than the one their text states. All limits were set before the results and none was moved, though one check's target was corrected (section 6): 215 checks pass and 38 miss. Every miss is listed with its cause or marked open. The largest are climb after a motor failure with distributed thrust, 86 against 360 ft/min, the climbs with both cruise motors running, 70 to 170 ft/min above the simulator's, and three frequencies against the Mod II simulator.

1 Introduction

The X-57 Maxwell was a battery-electric demonstrator built on a Tecnam P2006T [1]. Mod II replaced the two piston engines with electric motors in the same places. Mod III replaced the wing with one of 42 % of the area, aspect ratio 15, and moved the cruise motors to the wingtips, where their propellers turn against the tip vortices. Mod IV powers twelve folding high-lift propellers along the leading edge, which Mod III carries folded and inactive [1], to bring the stall speed back to the P2006T's [2]. The project ended with Mod II built and tested on the ground, the Mod III wing built and vibration tested [3], and neither of the later forms flown.

That leaves an unusual problem for a flight model: a large and open body of data, with no flight to hold it to. The CFD database [1] covers the wing, flaps and propellers; the high-lift propeller was tested in a wind tunnel [4]; the batteries, motors, controllers and cockpit are documented [7, 8, 9]; and the piloted simulator's results are published [11, 12, 13, 14]. All of those except the tests are predictions. Holding one prediction to another shows only agreement, so each check here is labelled by what it is held to: **M** measured, **C** CFD, **S** a published simulator, **D** a design document.

A table model on the database's own structure reproduces the CFD by construction, and is kept as the reference. It can't lose one motor's slipstream from one part of the wing except through the database's increments. The model that ships is built up instead: strips, discs, tail and bodies, whose failures come out of the parts.

2 Reference data

When sources disagree the shipped model follows the highest-ranked: flight test of that aircraft with a clean method, then wind tunnel at a comparable Reynolds number and configuration, then CFD, then a published simulator, then handbook estimates. A simulator ranks low because it's built from the others. The model is never tuned towards two sources at once. A comparison with a simulator is run on that simulator's inputs (its mass, its trim, its configuration), so that what's left is the model's own error; a part that vanishes was only a difference in data. Rank doesn't excuse a source from a sanity check: section 5.1 sets aside a higher-ranked figure for a stated reason.

3 The model

3.1 Implementation

One generator writes the Mod III, Mod IV and P2006T builds as plain JSBSim XML, with no scripting in the flight model. Forces and moments go through the aerodynamics axes. The build runs at 120 Hz, a rate at which the bending modes, flights and failures have converged. Inside FlightGear the force graph has to run as a system channel with stored outputs: read through FlightGear's property tree it would be worked out again at each read, and the start never finishes. Every property read in FlightGear takes a lock, so the shipped graph is laid out to read few: functions read once are written into their reader, unread ones are dropped, and the high-lift channels rest while those propellers are stopped and disarmed. That takes Mod IV from about 6900 reads a step to 6600, or 5600 at rest. A check flies this build beside the plain one, and the two match to the last bit.



Figure 1: The model in FlightGear: Mod IV in a banked turn with the high-lift propellers running.

Table 1: Sources and what they contribute.

Source	Content used
TM-20250001715 [1]	CFD database, power off and on, ground planes, contactor failures, model structure
NASA OpenVSP model	geometry of wing, nacelles, blades
20210016834, 20220006506 [4, 6]	high-lift propeller in the tunnel, blade weights, wake survey
20240006496, 20230004174 [11, 12]	simulators: mass, modes, failures, Mod II's identification
20240006682, 20240010539 [13, 14]	motor failure histories, DiTTo climbs
20240011864, 20240007439 [3, 15]	Mod III wing vibration test, flutter model
ICD-CEPT-006 [7]	cockpit, switches, sequences, displays
20240009560, 20240005892, 20170005797 [9, 8, 10]	motors, controllers, power system
Chin et al. [17]	cell model, thermal
P2006T flight manual [19]	performance, stall, gear system
Nicolosi et al. [22]	P2006T flight tests, tunnel, derivatives

3.2 Airframe

The wing is 17 strips a side (fig. 2): 10 flapped, 4 with the aileron and 3 plain. Strips end at each propeller's edge, so that a disc's slipstream has strips of its own. Their sections from NeuralFoil on the geometry's own airfoils and a lifting-line solution for the spanwise loading. The fuselage's crossflow, a circle's doublet, reaches the strips beside it, which gives the high wing its dihedral effect; the strip over the cabin, inside that circle, carries the fuselage's carry-through lift as wing, so the loading runs on across without root vortices. The slotted flap's section lift, moment and drag are fitted to the CFD at flaps 0, 10 and 30, power off, and flaps 20 is left out of the fit as a test. The aileron is NeuralFoil's plain flap, what each deflection adds taken at the section's zero lift and held across the attached range, with no gain: read at the strip's own incidence it faded where an up aileron's lifting line lifts its strips 7 to 8°, while the CFD's aileron is linear from −25 to +5° and nearly the same at 2 and 8° [1]. Past that, toward the stall, the aileron's lift is the plain section's shifted by each deflection's equivalent angle, as the slotted flap's is, and keeps at maximum lift the share of its increment that NACA's plain-flap tests keep: about 0.4 down and 0.8 up, from the 23012 with a 0.20c flap in the variable-density tunnel [30]. NeuralFoil's own up-deflected section climbs twice as steeply as the plain one from 8 to 14°; those tests' up flaps keep the plain slope. Tail, fin and bodies follow the method proved on the P2006T. The wing's downwash at the tail comes from six lumps of its lift, three a side: inboard, the flap's outer part from an aileron's span inboard of its end, and outboard, each shaped as the plain wing's loading inside it, and each averaged across the tail strip it reaches. In a sideslip the wake runs back along the air, not the body, and lies beside the tail: about 1.7 m at 20°, half the tail's span. Two lumps, inboard and outboard of the flap's end, let a right aileron's lost lift and the loss its lifting line spreads inboard all but cancel at the tail. Strip by strip is no better, since this straight, rigid wake carries a blown flap's and a slipstream's edge vortices at full strength past the tail, where the real ones roll up and spread. Past 8° a pitching moment on body incidence carries what the strips can't, because the CFD's pitch break comes at the same incidence with the propellers on or off, where a blown section is still short of stall. Rate derivatives are carried from the strips' to the tunnel-based ones the simulator uses, because without that the roll modes run 19 to 34 % quick. The correction is taken on the tunnel's curve a degree apart and, from 10°, every quarter degree, since the strips stall one at a time. With the high lift on, that correction holds its value from 2° under the power-off stall: past it, it stands for unblown sections stalling, which blown ones don't.

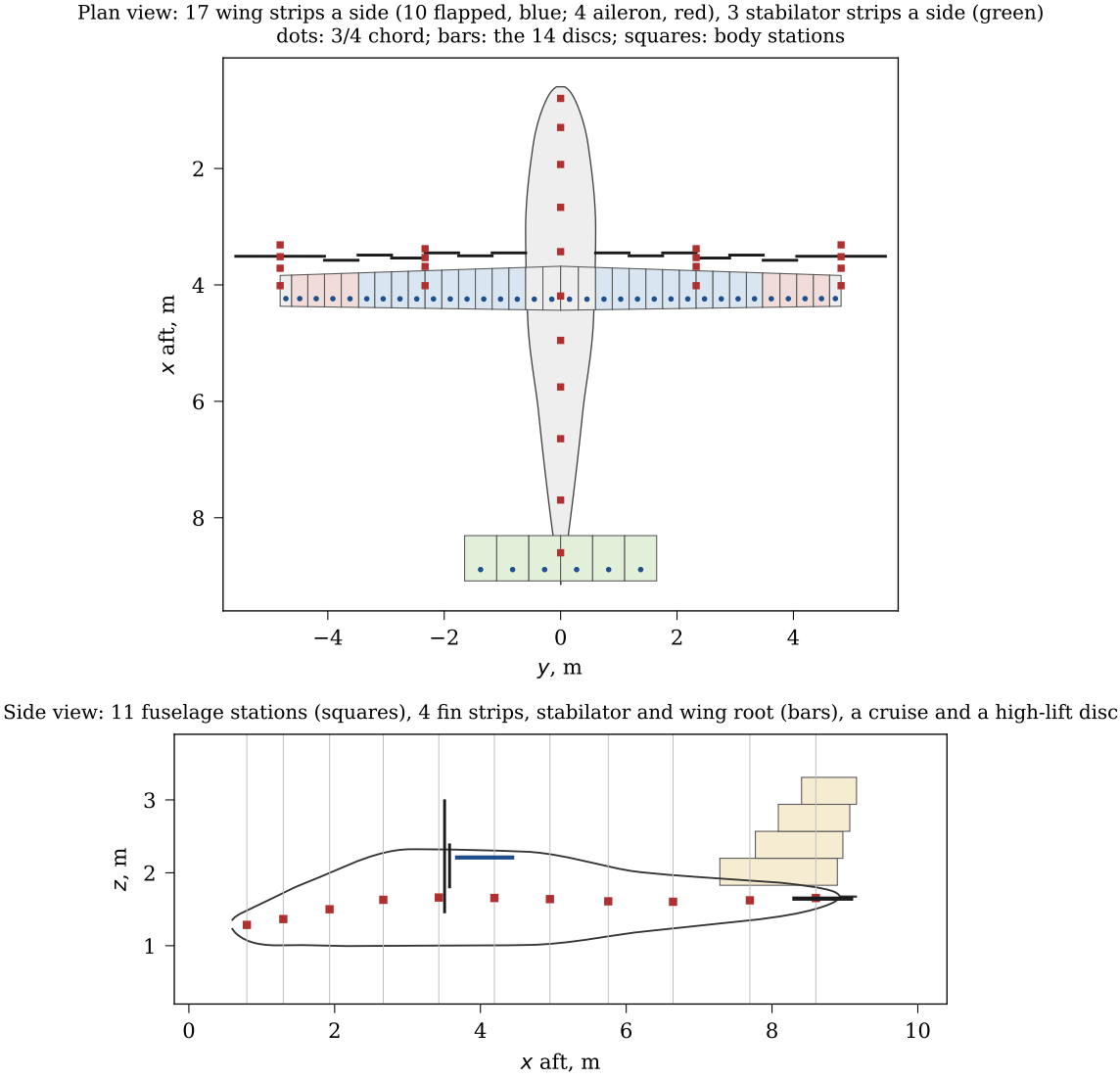


Figure 2: The model's discretisation, drawn from the generator's data, Mod IV. Strips end at each disc's edge, and at the flap's and aileron's. The P2006T's build has 13 wing strips a side (6 flapped, 3 aileron, 2 on the winglet), with the same 3 stabilator and 4 fin strips and 11 fuselage stations. The tip nacelles have 4 stations each; the high-lift nacelles are lumped, six to a side at their mean span, also on 4 stations.

3.3 Propellers and motors

Propellers are discs, not blades: the blades of twelve propellers pass at 400 Hz, far beyond the frame. Each of the fourteen discs keeps its own speed and takes the air at its own place, the airplane's roll, pitch and yaw included: a yaw rate speeds one wingtip's disc and slows the other's. Thrust, torque and in-plane force come from offline blade-element tables on the published blades. The high-lift blades run about 10° folded forward on their hinges, so the tables use an effective 20.0° against 27.35° drawn: partly documented, partly fitted to the published schedule. The cruise propeller's speed is governed by its blade pitch, which an electric hub changes at 1°/s, and each motor is a torque balance on the rotor's inertia. No rotor inertia is published. The cruise rotor's 0.65 kg m^2 is about 0.5 for the propeller, from the manufacturer's 1997 data sheet (0.6 kg m^2 with 160 cm blades, scaled to 152 cm), and 0.15 for the motor's rotor, estimated; the range is 0.58 to 0.73. The high-lift rotor's 0.0085 comes from the measured blade weights and the reference motor's parts [16].

The cruise propellers turn against the tip vortices, inboard blade rising: the published statements [28, 26] and the published 3-D geometry, measured blade by blade, agree. Each wing's six high-lift propellers turn the same way as its cruise propeller, the wings mirrored. The published geometry builds them so. The Mod IV CFD's skin friction [26] has the same lopsided pattern in every bay of a wing and its mirror image on the other, where alternating propellers would flip it from bay to bay. And on the X-57's predecessor wing, CFD gave propellers all turning against the tip vortex 0.31 more lift coefficient than pairs turning towards each other [27]. The simulator paper's "each pair of motors was counter-rotating" [11] is taken as left against right. It can also be read as neighbours, as this model first did, so that simulator may differ.

3.4 Slipstream

The axial speed at each strip comes from a tube of ring vortices behind the disc, after Heyson and Katzoff [25], the model with which the same author's V-22, UH-60 and Bo 105 take each rotor's downwash to the airframe. Their rotors' own inflow is a Peters–He dynamic wake; here the disc tables give it, in steady flow. The tube's contraction, 0.90 of the radius, is momentum theory's. The slipstream's effect on a wing is then reduced by one fitted constant, on the jet's strength:

$$a_{\text{eff}} = a\left(1 - k\,\frac{a}{u+a}\right), \qquad k = 0.70 \tag{1}$$

It probably stands for a jet no taller than the chord, which does less than strip theory says; that isn't proved. The plain momentum jet this replaced needed three fitted constants for much the same result (section 5.5). In a sideslip each jet drifts with the crossflow, the less the stronger it is: per foot behind the disc, $-v/(u + 1.5a)$ over the wing and $-v/(u + 2a)$ at the tail, much as the same author's UH-60 carries its main rotor's wake to the tail by speed and direction. Each strip's share of each jet is a table on that drift, nil at none, so straight flight is as before. At 10° of sideslip it gives the CFD's growth of power-on roll as speed falls: −0.035 against −0.048 at 50 KCAS, from −0.024. At 20° the CFD's roll grows faster than any drift gives. No dynamic inflow is modelled: a Peters–He wake on each disc would give its own inflow, which the tables already hold in steady flow, and wouldn't change how the slipstream reaches the wing.

Each jet also turns, its swirl carrying the propeller's torque as angular momentum. The cruise propellers' loading is published: NASA's CFD gives them Goldstein's [1], so their swirl follows the optimum's profile [24], a solid body's inside $r/R = J/\pi$, a free vortex's outside it and nil at the tip (Prandtl's loss for three blades standing for Goldstein's), from the nacelle's surface out, scaled to the torque and spun up as the jet contracts. That puts 1.7 to 2 times an even swirl on the strip beside the nacelle, across J 0.6 to 1.6; the model takes $J = 1$. The swirl also turns the flow each strip's force stands on, so a strip in its upwash leans its lift forward: the slipstream's rotation won back as thrust, the reason the cruise propellers turn against the tip vortex. Behind the high-lift discs the swirl stays even from 0.35 of the radius out, a stand-in for an unpublished BEM loading, and leaves the force's direction alone: with wing on both sides of a hub the net lean is the difference of two large ones, and given the even swirl the high-lift drag fell further under the CFD's, 0.082 against 0.119 at flaps 10 from 0.106. With one cruise propeller at the CFD's thrust, as the CFD runs it, 41 cases from 43 to 142 KCAS, the model gives 1.07 of the CFD's added lift, 1.01 of its drag reduction and 0.88 of its pitch, rms errors 15, 48 and 21 % (the drag's 0.0012); the strips' own roll is 1.03 of the CFD's. With the swirl even and the force not turned it gave 0.73, −0.99, 0.39 and 0.71: drag rose with thrust where the CFD's falls. At 150 KTAS the two now take 0.0023 off the cruise drag coefficient, where NASA's report puts the tip propellers' benefit at 0.0028 [1]. Nothing in this was fitted. The side force, nil in the model, and the yaw are under the 0.005 floor of the limit, as is the drag.

3.5 Named fits

Fits are named, and a fitted figure can't pass a check by itself. Besides the sections and those above: the flap's turning of the jet; the slipstream at the tail; a jet's lift from nearer the centre after a contactor failure; sideslip's roll and yaw on sideslip, incidence and flap, from the CFD, power off, because the strips lose the windward lift near stall where the CFD keeps it, with the high lift on only its part at 2°, far from stall; one propeller's roll from the published tables, less the strips' own share, because a windmilling propeller takes lift off its own wing, which the strips, fed by thrust, don't (powered, the strips now give the tables' roll on their own); the rudder, whose section table ran out towards full travel; ground effect on height, incidence and flap, power off, and with the propellers on per unit of their thrust coefficient; the gear's drag area of 4 ft^2, from the simulator's phugoid; the brakes' force, from the P2006T's landing roll; and the nose wheel's play, 2° either side of the pedals' angle, from NASA's straight-line steering in a 15 kt quartering tailwind (section 5.8).

3.6 Power, battery and cockpit

The traction battery follows a cell model [17]. Seven avionics buses, the essential battery, two converters and their cross buses run inside the flight model, so that a shed bus or a failed converter reaches the contactors and the motors as the handbook says [7]. Voltages are fixed and loads estimated. The cockpit's logic is in the aircraft's own files; the companion manual describes it.

3.7 Wing bending and gear

The wing bends in its first mode, symmetric and antisymmetric, as in the same author's ASK 21: stiffness fitted to the Mod III wing's vibration test [3], with tip masses and without, each nacelle a mass with the rotary inertia of its size. The panels' coupling to the airframe is solved in the system, with an effective mass for each kind of motion. Tyre deformation drag follows the published equation [18], held at its value past 55 KIAS, where its fit ends. The brakes are the P2006T's, which the X-57 kept: a disc gives a torque for its pressure, so the pedal sets a force at each main tyre, capped by the tyre's grip. Their force at full pedal, 237 lbf a wheel or about 0.2 g with both, is fitted to the manual's landing roll. Each tyre's side force follows Smiley and Horne's measured cornering power for its size, load and the manual's pressure [20, 21, 19]: 0.112 of its load a degree for the nose's 5.00-5 and 0.061 for the mains' 6.00-6, against JSBSim's 0.121 for every tyre, on JSBSim's own curve to the same peak of 0.72. The pedals steer the nose wheel in proportion to 20°, as NASA's ground simulation from Tecnam's data has it [18], with slack in the linkage that NASA's pilots noticed; within 2° of play either side of the pedals' angle the wheel trails, taking no side force. The play isn't published and is fitted to NASA's simulator (section 5.8). Gear terms reach the panels as the mean of two frames, which removes a 60 Hz shake between JSBSim's tyre friction and the panels' reaction.

3.8 Mass

FlightGear flies the published gross weights: 3000 lb for Mod III [1] and a projected 3200 lb for Mod IV [15]. Comparisons

Table 2: All checks, by what they're held to: M measured, C CFD, S a simulator, D a design document.

Checks	Against	Pass	Miss
Modes and CFD increments, built-up model (limits written for the table model)	C, S	2	10
Propellers, motors, governor, battery	M, S	26	3
P2006T: flight manual, flight tests, tunnel, Mod II simulator	M, S	48	16
X-57 built-up aerodynamics against CFD	C	37	2
Ground effect, climbs, failures, gear	C, S, M	44	6
Wing bending	M, S	31	1
Cockpit power and propeller modes	D	27	0
All		215	38

with the simulator run at its 3075 lb. The 28.6 % CG and the inertias are the simulator's [11]: no other is published. With no fuel, weight and CG stay constant through a flight.

4 Method

The work went in phases, each with limits written before its first result, taken from the sources' own spread where they have one. A limit isn't moved afterwards, and a miss stays printed until its cause is fixed. One script refits the model and runs every check, at four processes. Where a test's configuration turned out to be wrong it was changed and the old run kept printing beside it, with the reason given. Table 2 gives the count by group.

5 Results

5.1 The method on the P2006T

(M) The built-up method passes the P2006T's flight manual: 27 cruise points mean +1.3 kt, rms 1.7, worst 2.7 (fig. 3, one drag fit); climb within limits on two engines and one, 1129 against 1102 ft/min at maximum weight and 217 against 206 on one engine. Stall speeds are fitted. Against the tunnel, part by part, it gives lift slope 0.0825/deg (0.080) and the neutral point 41.4 % (41). In flight, phugoid 28.8 s (27), damping 0.088 (0.09); Dutch roll 2.96 s (3.25), damping 0.305 (0.26).

Six checks against the flight paper [22] miss, and a second source says why. The simulator for Mod II was built on an identification of an instrumented P2006T and shown matching that aircraft's flight data [12], and its paper prints mass, CG and inertias, so the model can be run at them **(S)**. Its short period then agrees across the range (fig. 4): 3.85, 4.47, 5.13 rad/s against 3.7, 4.4, 5.2, damping 0.53 to 0.56 against 0.59. The flight paper's 3.41 rad/s and 0.40 are the outliers. Its maximum-slope method needs a true step, and the pilot's took about 0.6 s: the same method on the model's response to a 0.6 s step gives 3.68 rad/s, though the model's root is 5.92 at that mass and CG. The lift slope misses too, but the paper's 4.15/rad is the outlier again. Read together, because they share their points, the simulator paper's figures 11 and 12 give C_L against α: 5.09/rad, straight through all five cruise points. The nose vane appears to read about 1.3 of true

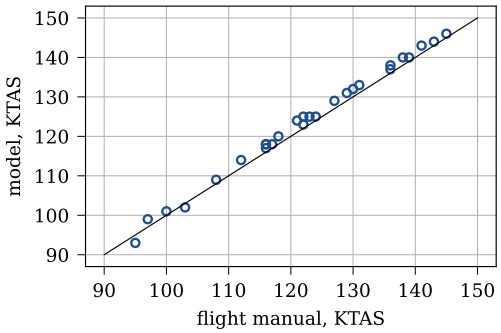


Figure 3: P2006T cruise speed at the manual's power, 27 points **(M)**.

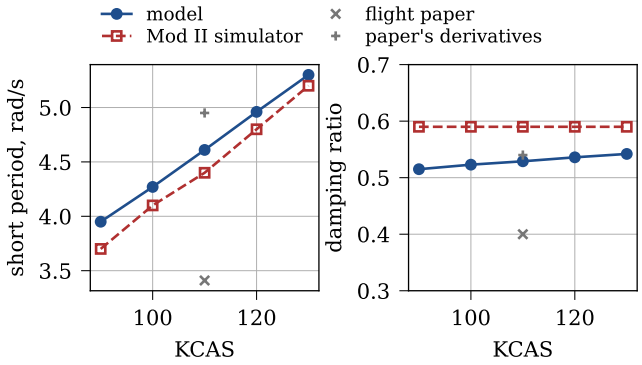


Figure 4: Short period at the Mod II simulator's mass against that simulator, read by eye **(S)**. The flight paper's measured figure and the one its own derivatives give, at 110 KIAS and its lighter mass and forward CG, are shown for scale.

incidence, and the simulator's authors replaced the published slope themselves [12]. The model's trimmed 5.70 is still 12 % above the simulator's, a real excess, and a miss. Aileron per degree of sideslip is 0.75 against the flight paper's 0.58 and, at the simulator's mass, 0.93 against its 0.80. The tunnel's $C_{Y\beta}$ of −0.69 stays open: sources run from −0.36 to −0.69, and the model gives −0.48, or −0.65 with winglets and power.

Against the same simulator three other frequencies miss. Dutch roll is 42 % high at 110 KCAS (1.80 against 1.27 rad/s), but scaling the flight test's 2.0 rad/s by the inertia gives 1.75, so measurement sides with the model. Phugoid is 17 to 24 % high, and the flight test's 27 s sides with the model again. The roll time constant is 0.1 s against the simulator's 0.7 to 0.9 s: any published C_{lp} at the printed inertia gives 0.1 to 0.2 s, and the flight paper calls the mode too short to observe. That gap is unexplained.

None of this reaches the X-57's lift or stability, which come from the CFD: the P2006T is where the method was proved.

5.2 X-57 aerodynamics

(C) Power off, the fitted flaps match the solvers' mean within 0.003 in C_L and 0.020 in C_m; flaps 20, unfitted, within 0.055 and 0.026. Unfitted increments below stall: stabilator C_L 16 % and C_m 7 % rms, aileron roll 14 %, lift 18 % and pitch 20 %, sideslip side force 8 %. The aileron's pitch is 0.80 to 1.05 of the CFD's up and 0.79 to 1.19 down, and at 8° and −25°, where the up aileron's strips reach 16°, its roll is 0.97 and its pitch 0.87 of the CFD's. That pitch error was 52 % while the aileron's lift and pitch carried gains of 1.5 and 2.0, before the section's increment was held at zero lift and the

tail's downwash came in lumps, and 30 % before the section kept NACA's share of its increment toward the stall. Near stall, at flaps 0 and 16°, 10 and 15°, 30 and 14°, the aileron's roll misses at 27 % against 25 (31 before): up, 1.14 to 1.41 of the CFD's; down, 1.5 at flaps 0 and 10 and 0.72 at 30. Two of the CFD's cases at flaps 10 are left out of that check: its lift rose with the aileron up, so its separation settled unlike the neutral case's. Sideslip's moments and the rudder are fitted now and are no longer tests, power off. With the propellers on they still are: 26 and 9 % rms of the increments in roll and yaw (limit 25), the roll a narrow miss, 23 before the lumps. Trimmed stall speeds for Mod III are 82.7, 80.6 and 70.8 KCAS against the published 88, 80, 73, worst 5.3, a narrow miss. At 150 KTAS L/D is 14.2 (13.5); the peak 15.6 (15.9) at 110 KCAS. One cruise propeller, against the power-on CFD: section 3.4.

5.3 Propellers and battery

(M) The high-lift disc's tables are inside the tunnel's scatter: C_T rms 0.0064, C_Q 0.0007. One radius behind the disc NASA's wake survey measured the slipstream at 58, 75 and 90 KEAS [6]. The model's, momentum theory on the balance's thrust over the whole disc at a fixed contraction, is 26 and 32 % slow at 58 and 75 KEAS and its edge 0.07 R in at 75: three misses. The measured stream carries 1.18 to 1.22 of the balance's thrust. The blades' own thrust is more than the balance reads, which takes the spinner's and nacelle's blockage with it [6], and NASA's blade-element table gives it; the swirl's low pressure adds about 4 %. Built that way, on the annulus outside the nacelle and narrowing by momentum theory, the jet comes within 12 and 16 % and every edge within 0.03 R. Its lift, through NASA's velocity multiplier for a slipstream of finite height [5], taken at the round jet's own height strip by strip, then strays from the CFD's: 0.14 to 0.21 under at 35 KCAS and up to 0.09 over at 75, rms 0.129 against the shipped jet's 0.085, and a cruise propeller's lift goes from 1.08 to 1.21 of the CFD's. The two-dimensional multiplier loses effect with the jet's speed faster than the three-dimensional CFD, and the old saturation refitted on the new jet can't follow the CFD's trend either (rms 0.104 at best, still −0.06 at 35 KCAS and +0.09 at 75). So the shipped jet stays a stand-in fitted to give the CFD's lift, slower than the real stream; the measured one is kept behind a switch. The windmilling disc turns 2640 rpm at 75 KEAS, inside the measured 2483 and 2654. **(S)** Torque is within 7 % of the published schedule to $J = 1.1$, the cruise tables pass 37 published points to 0.0003, and the rpm schedule is exact to 0.2 rpm. The published slow-flight mission draws 41.9 kWh (41.9), lowest 378 V (370), depth of discharge 80.0 % (82.6), hottest pack 59.3 °C (57.9). The cell is within 0.09 V of the published figure.

5.4 Flight modes

(S) Against the simulator's linear models [11], Mod III is within 0.012 in phugoid damping, 0.036 short period, 0.025 Dutch roll, 13 % roll time constant, and 10 % frequency (fig. 5). Mod IV is within 0.02 to 0.08 and 23 % roll. Phugoid frequency in the airspeed mode is 33 % off at 88 KCAS (0.26 against 0.19 rad/s), where the simulator's figure is off its own trend of 0.24 and 0.25. Those limits were written for a table model, and some still print small misses. In-plane propeller force, and each disc's own air when the airplane turns, cause all but that one. Both add damping the simulator doesn't seem to have: both are real, and they stay. The second adds 18 % to the yaw damping at 112 KCAS, 455 ft lbf s on the airframe's 2527, as $2y^2\,\partial T/\partial u$ gives by hand. The roll mode is quick against the simulator everywhere, by 11 to 13 % in Mod III and up to 23 % in Mod IV's airspeed mode at 75 KCAS, and NASA's own database is too: flown here as the table model at the simulator's inertia, that case takes 1.03 s against the simulator's 1.40, the built-up model 1.08. Three things make the gap. The propellers' in-plane force: the simulator's propulsion takes only thrust and torque [11], and the CFD's discs had prescribed loading, so neither has it; without it the table model takes 1.11 s, and in Mod III comes within 2 to 5 % of the simulator. The blown wing's added damping, which NASA's powered forced-oscillation CFD shows and the built-up model follows: without that too, 1.22 s. What is left, 13 % at 75 KCAS falling to 5 % at 96, the published data don't explain. The simulator rates Mod IV's roll borderline Level 2 at 73 KCAS [11]; this model's is quicker there. At 80 KCAS, Mod III's stall speed, Dutch roll falls to 0.06 against 0.11. The table model gives 0.06 there too, so that follows the CFD's loss of directional stability, not the built-up model.

Table 3: Slipstream models at 58 KCAS, flaps 30.

	Constants	$C_{L\,\max}$	C_L error
Limit		4.40–4.58	0.078
Jet, as first fitted, discs alternating	3	4.52	0.078
Jet, no fit	0	5.55	0.690
Ring wake, no fit	0	4.99	0.327
Ring wake	1	4.54	0.072

5.5 High lift

(C) With the CFD's thrust, Mod IV at 58 KCAS and flaps 30 reaches $C_{L\,\max} = 4.53$ (4.40–4.58), the propellers adding 1.23 to 1.63. Figure 6 and table 3 show the three slipstreams tried. With no constant at all the ring wake halves the plain jet's error and brings moment and drag to their limits. Lift stays about 0.3 high, and one constant takes it inside the solvers' spread, as well as the three it replaced.

5.6 A motor lost

(S) The published failure, no pilot, 6000 ft, 95 KCAS [13], is described as takeoff flaps and gear down. Its traces can't be that. Its trim incidence of 8.4° and Mod II's 5.4° both sit on the cruise line of the Mod II paper's figures, and its airspeed falls at the rate of about 115 lbf lost, where takeoff flaps and gear need 180 a motor. The table model, flown clean, lands on every trace at once (fig. 7). This is an inference from the published numbers, not something stated. The test now flies clean, with its targets and limits as they were, and the takeoff-flap run still prints.

Flown clean, the built-up model at first rolled too little, −60° at 3 s where the table model gave −81°. That was a real departure from the CFD, in two parts: sideslip's roll near stall, and one propeller's own roll. With both fitted it gives bank −74°, pitch −9.9°, heading −23.2°, peak roll rate −46 deg/s

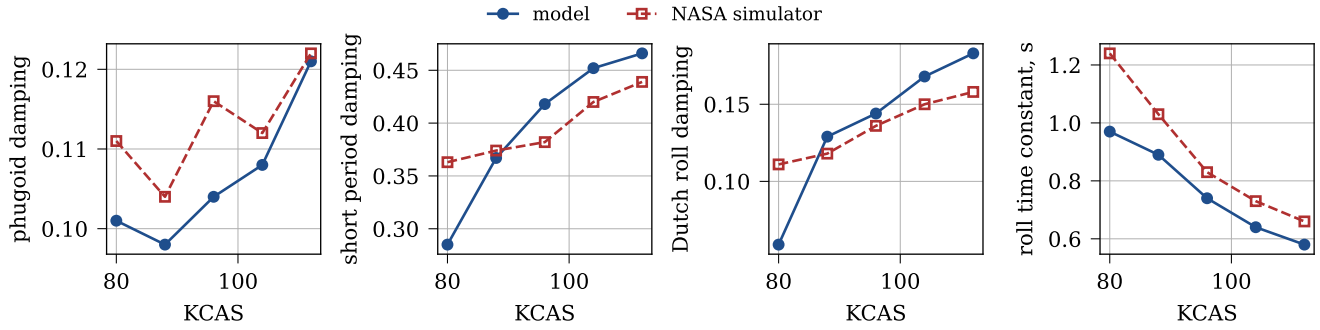


Figure 5: Mod III's modes against the simulator's **(S)**.

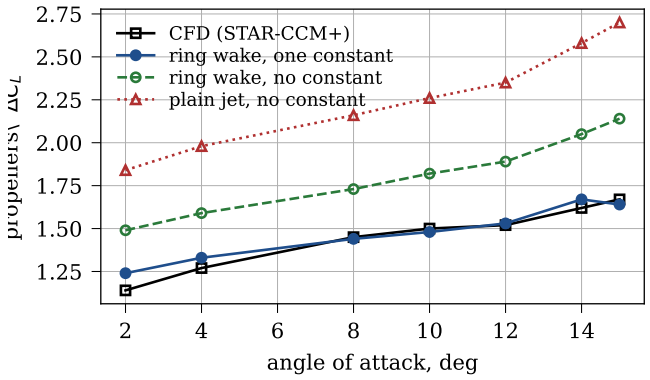


Figure 6: Lift the high-lift propellers add, with the CFD's thrust **(C)**.

and sideslip 10.5°, sink 18.7 ft/s, against about −87°, −11.5°, −23°, −47, 9.1°, 24.5. The check that first flagged the sideslip as twice the published figure was comparing unlike configurations.

5.7 Climbs, control and failures

(S) With a motor out and feathered Mod III sinks 418 ft/min (450), and 315 (300) with a traction bus out. Minimum control speed is 84.2 KCAS (85), set by the rudder, which travels 26° each way as on the P2006T [19], in NASA's ground simulation [18] and where its Mod III pilots hold it [13]; the CFD swept it to 28, its tolerance's far end, and with 28 it was 83.3. Every row of the published failure list [11] is present and does what the row says. A contactor failure at 58 KCAS costs ΔC_L −0.30 and −0.34 (read as −0.30 and −0.35) **(C)**.

Six climbs miss. Mod III at full power climbs 597 ft/min (430), and at 2400 rpm 426 (270); DiTTo at 84 KIAS 1151 (1000), and with a contactor out 599 and 893 (480 and 740). With both cruise motors running every climb sits 70 to 170 ft/min above the simulator's; the cruise propellers' drag reduction from the CFD (section 3.4) added about 80 of that, and 20 to 40 with one out. NASA's database holds the same increments, each side's on J and blade pitch [1]. The rest is thrust. Clean, against NASA's performance estimates, made with its simulator's own models [23]: with no thrust the model's L/D at 8000 ft is within 3 % of NASA's from 120 to 200 KTAS (14.9 against 14.8 at the peak, 13.6 against 13.3 at 150 KTAS), with more drag below that, yet at maximum continuous power and 2500 ft it climbs 785 to 826 ft/min at 100 to 110 KTAS against NASA's 600, and 90 to 110 more from 137 to 163 KTAS. The thrust NASA's figures imply is about 3 % under the model's at $J = 1.35$ and nearer 18 % at 0.9. NASA's cruise propellers carry an installed loss matched to J from Mod I's flight-tested thrust [23], not published; this model has the uninstalled tables both start from. That loss isn't fitted: its size isn't known, one that swings from 3 to 18 % reads more as compensation than as an installation, and a flat 8 % that brings the climbs with both motors inside their limits takes the bus-out ones outside. Nor is it the slipstream counted twice: in the climb the cruise propellers' wake takes off half to two-thirds of the CFD's drag reduction, and with no wake at all the model still climbs 45 to 130 ft/min over NASA's figure. DiTTo with a motor out gives 85 and 453 ft/min (360 and 530). The published drop at 97 KIAS implies 108 lbf lost from a motor whose published tables give 173, and the table model drops by the same as the built-up one, so this isn't the built-up model's. The arithmetic is short: climb lost is thrust lost times true airspeed over weight, so about 170 lbf net at 170 ft/s on 3075 lb costs some 560 ft/min before any rudder and aileron drag. The model loses 643, NASA 360. On the real P2006T one engine out costs 81 % of the climb [19]; here it's 90 % at 97 KIAS and 59 % at 84, against NASA's 50 and 47. That comparison is loose, since a tip motor is about 30 % of the thrust and not half, and NASA's simulator may hold an effect its published tables don't show. DiTTo with a bus out gives −174 and 87 (−100 and 100).

(C) Ground effect is fitted power off, so it tests nothing there. With the propellers on, the model's own ground effect, the wake's image on the wing and at the tail, is within 0.03 in C_m with takeoff flaps and, with landing flaps, from 65 to 90 KCAS; it falls short as the blowing grows, by 0.28 at 35 KCAS. There the CFD's change keeps a pitch-to-lift ratio of about −5 at every speed, so it acts at the tail, while near the ground the model's blown wing gains two to three times the CFD's lift. The image of the blown loading doesn't explain it: lump by lump the tail's relief is 0.41 to 0.44, the plain wing's 0.44, and the wing's own grows. It reads as the slipstreams, turned down by the flaps, meeting the ground, which the model doesn't have. What's left is fitted per unit of the high-lift discs' thrust coefficient, on height, incidence and flap, from LAVA's ground planes, the one solver that ran them: worst 0.110 in C_L and 0.136 in C_m against 0.15, rms 0.017 and 0.039. Averaged over the speeds, as it was, it moved the slow cases' error onto the fast ones, where the model was already right: worst 0.268.

5.8 On the ground

(S) NASA's open-loop steering runs [18]: a steady ground speed, then at once a nose-wheel angle, the rudder in propor-

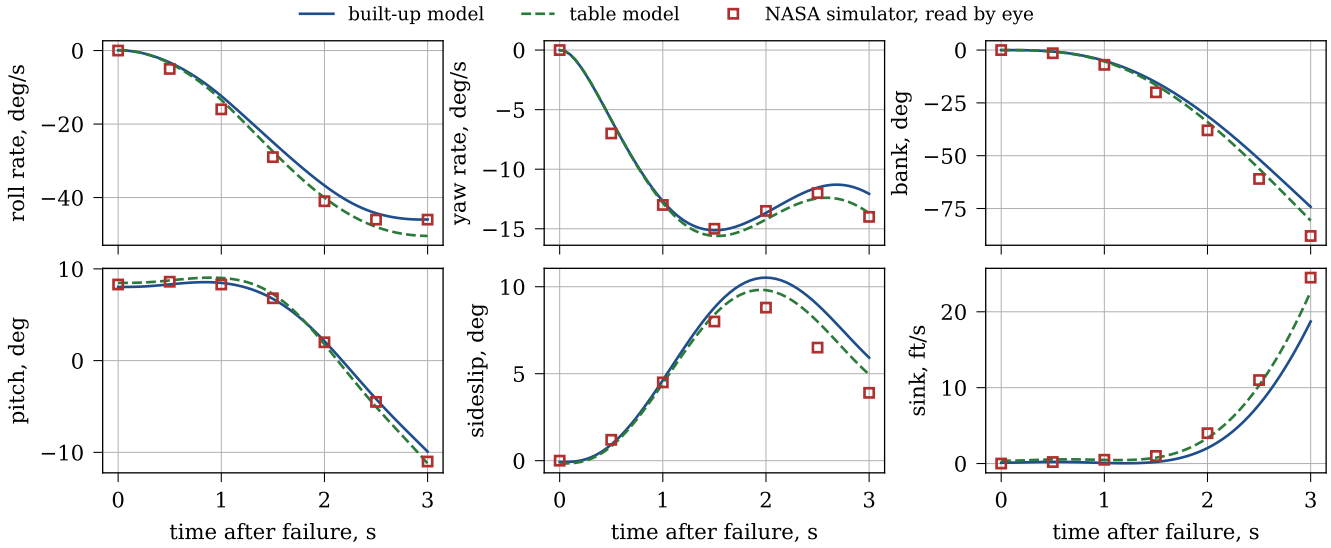


Figure 7: The left motor lost at 6000 ft, 95 KCAS, no pilot, flown clean **(S)**.

tion and a 15 kt wind from 45° behind on the right, elevator and ailerons neutral. At 10 kt the turns follow the geometry, 24 to 26 ft at 20° (26), and at 5°, of which the play can take 2, 121 to 195 (108 to 181). A slow heading hold on the nose wheel finds the angle that keeps a straight line: with takeoff flaps 2.5, 3.0, 3.2, 3.3, 3.3, 2.9 and 2.0° from 20 to 80 kt against NASA's 3.7, 3.2, 3.2, 3.2, 3.2, 3.0 and 2.0, which is what the play is fitted to; 1.9° rms off with none. With landing flaps it gives 2.5 to 3.5° against NASA's 4.5 to 5, and the P2006T's own model at Mod II's mass, NASA's aircraft, matches at no play.

Inputs of 5° or more from 20 kt lean the aircraft onto a wingtip propeller, as a rigid nose wheel on gripping tyres must: 5° at 30 kt asks 0.7 g by geometry, and this gear tips near 0.5 g. NASA's simulator, with the same grip of 0.72, doesn't tip below rotation speed even at full nose wheel, and its turning radii at 10 kt are 20 to 50 times the geometry's, which no tyre's slip explains, so its open-loop map isn't compared. With the pedals held the aircraft is directionally unstable at speed: the nose tyre, 0.112 of its load a degree, works 8 ft ahead of the centre of gravity, the mains, 0.061, 1.4 ft behind it. The play keeps it near the centre, and with no input in that wind it drifts into the wind as NASA's does, except at 40 kt with takeoff flaps, where it still leans onto the propeller. Past the play it is the pilot's: NASA's pilots found skidding with aggressive inputs from 60 kt [18].

5.9 Wing bending

(M) Fitted to the vibration test's first modes (2.25 and 6.55 Hz, 2.14 antisymmetric). The nacelle's rotary inertia about the fore-and-aft axis is estimated from its size, not fitted: its 134 lb as a solid cylinder filling its widest section, 0.93 kg m^2. Unfitted: with the tip masses, second bending 17.00 Hz (15.76) and 15.90 (14.82); clean, second bending 27.78 (26.36) and 25.76 (23.90), and antisymmetric first bending 6.06 (5.76), 5.2 % high against a limit of 5, a narrow miss. Without the nacelle's inertia, second bending with the tip masses was 18.50 and 17.06 Hz, 17 and 15 % high, because that mode turns the tip most. The model only flies the first mode. **(S)** Whole aircraft 2.39 Hz (2.31) for Mod III and 2.37 (2.29) for Mod IV. In JSBSim with the air nearly gone the modes ring at the beam's frequencies within 1 %; in flight they're damped at every speed to V_{NE}. There's no torsion, so no flutter prediction. Bending off gives the rigid model's forces exactly.

5.10 Cockpit and FlightGear

(D) The avionics buses, sheds and propeller modes pass against the handbook. In FlightGear each overhead and lower switch, the wired breakers, light tests, alerts, manual pitch, the lever's feather stop, the high-lift logic, flaps, Autostart and Autostop were driven by the property each control sets. All 113 clickable controls have their objects and Nasal. Hands off in cruise it holds height within 3 ft in 60 s. Real mouse clicks were not automated.

5.11 Sound

Every sound plays on one scale, a full-scale sample at volume 1 being 112.5 dB re 20 μPa, in the cockpit and outside alike. The propellers, motors, the wind on the windscreen and the gear in the airflow are computed by PyWopWop [31], the last by Fink's method [29] as for the ASK 21's fixed wheels: at 90 kt the gear is 66.7 dB(A) at 30 m, about one cruise propeller's level, and 73.3 in the cockpit, some 16 dB under the wind. So are the tyres rolling, each wheel on its own speed: scaled from a measured truck tyre, outside they come within 0.5 dB of the one light aircraft measured (60.5 against 60 dB(A) at 30 m), and in the cockpit their level comes from a recorded landing in a P2006T's cabin, set against the wind: three wheels make 86 dB(A) at 30 kt and 89.5 at 60, perhaps up to 3 dB loud. The touchdowns are PyWopWop's too, each wheel's on its own. The flight model turns each wheel: on the runway friction spins it up, the tyre's sliding grip on its strut's force, until it rolls; in the air its brake's pads and its bearings slow it, 3 N m a main wheel and 0.5 the unbraked nose, estimates with no measurement behind them, which stop a main wheel from 85 kt in some 14 s, and the pedal stops it at once. As a wheel's weight comes on, the flight model holds its sink rate and its slip, the ground speed less its rim's. The tyre slides until it rolls, losing $I(slip/r)^2/2$ whatever the load, and the strut sets how fast the load comes,

its damping slowing the sink while the wing sheds the weight, so a firm touchdown squeals short and a gentle one long, the squeal following the load as the recorded one does; a wheel back on the runway after a hop is still turning, and quiet. The share of that loss heard and its spectrum come from the same P2006T landing, the nose tyre's spectrum from a glider's: at 60 kt of slip a main wheel's squeal is 83 dB(A) of sound exposure in the cockpit and 68 at 30 m, the nose wheel's 79 and 67. Each wheel's files are one waveform under different envelopes, cross-faded by amplitude over the slip and the sink rate, so a touchdown between them is one event at the law's level, within 0.2 dB, the energy their differences cost put back. The strut's thump, the force its damping puts into the airframe, was at the edge of hearing in that recording, so its estimate is uncertain; it plays at it, a main wheel's from 69 dB(A) at 0.25 ft/s to 81 at 1 ft/s, a firmer one through a soft knee where it would pass full scale, 3 dB under its estimate at 2 ft/s and 7 at 4. Those two recordings are the only ones used. Mods III and IV were never heard, so the rest is computed, not checked; the motors' levels and the wind's 91 dB(A) at 110 kt are the least certain. The gear's pump and locks and the flap motor use stock samples at stated levels, all estimates: none has a measurement behind it on this aircraft. The aural alert is set 15 dB over the cockpit's noise at 100 kt, after ISO 7731. A check reads each level back from the sound file, and refuses any condition FlightGear wouldn't load: one did, silencing the tyres until a recorded landing showed it.

6 Corrections found during validation

- A blade-element solver took the first root it met, and lightly loaded sections have false ones. Phugoid damping, which hangs on how thrust moves with speed, exposed it.
- The rudder scale sat ahead of a table ending at 26°, stopping it at 17°. Later the fin's table itself proved a quarter weak at full travel.
- A wing's divide by a speed near zero, and JSBSim's $b/2V$, threw the aircraft into NaN at rest with the rotors turning. Both are floored at 30 ft/s.
- Reading JSBSim's accelerations for the panels' coupling gave wrong frequencies and negative damping; it's solved in the system now. Raw roll rate fed to the ailerons pumps the 3.55 Hz mode up, so a test pilot's rate passes a 0.25 s lag.
- The force graph couldn't start in FlightGear until it became a system channel.
- FlightGear's default electrical system was running beside the model's and overwriting one voltage. It's off.
- The door handles called a module the package didn't load.
- The high-lift propellers first alternated along each wing, from a loose reading of the simulator paper. Turned as the geometry and the CFD have them, maximum lift rose by 0.06 with no constant changed, a contactor failure's yaw came inside its limit, which had been a miss, and the propellers' sound outside moved by up to 10 dB, because the side view hears the rising blades.
- The cruise blades in the visual model were pitched for the opposite rotation to their animation.
- The failure test ran the wrong configuration (section 5.6), and minimum control speed was read from an unsolved point.
- The gear retracted on the ground. The flight manual shows only a knob to pull first, and other sources differ on a squat switch. A guard was added as a simulator choice.
- The parked check asked for wings level, though the published CG sits 1.19 in left of the centre line [11]. The mains' springs and the CG's height give −0.17° for that, which the model shows; the check now asks for the statics' lean, within the same 0.1°.
- The brakes stopped the P2006T at the tyre's grip, 0.7 g, in 138 m against the manual's 209. The manual's table is for grass and paved is only 2 % shorter, so its brakes, not the tyres, set the stop. Given a force fitted to the paved roll, the same landing on grass, at the published grip of a tyre there [18], stops in 208 m against the manual's 213, an unfitted test. The X-57's brakes now hold it to half lever at 2700 rpm, and its takeoff test starts rolling, as the published technique does [14].
- The discs took the air at the centre of gravity, so the airplane's turning changed neither their thrust nor their in-plane force. Each now takes its own: the wingtip discs add 18 % to the yaw damping, and Mod IV's fixed-mode check, which passed, now misses the simulator's Dutch roll damping, by 0.085 against 0.05 with the sideslip fit below.
- With the high lift on, the power-off fit for sideslip's roll and yaw was switched off whole, on the ground that blown sections don't stall. At 2°, far from stall, it still holds the high sideslip's yaw, +0.013 at 20°, flaps 30, which the CFD with the propellers on shows too. That part now stays: power-on yaw from sideslip went from 16 to 8 % rms, roll from 26 to 23, inside the limit.
- The rates' correction stayed whole with the high lift running, and near stall it stood for unblown sections stalling, so blown roll damping was counted twice: −1.19 at 12°, flaps 30, 58 KCAS, where the strips alone and the CFD's forced oscillation with the propellers on both give −0.84. Held below stall, the model comes to the powered forced-oscillation CFD within 0.074 rms in C_{l_p}, from 0.222, and 0.024 in C_{n_p}, from 0.061. Those runs were flown at a fifth of the power-off runs' frequency, so comparing the two, as the CFD's own report does, mixes frequency with power; at one frequency, thrust adds no yaw damping to the airframe in either the CFD or the model, and roll damping in the model grows half as fast as in the CFD.
- The strip over the cabin sat inside the fuselage's circle, where the doublet doesn't hold, and its clamp left it 14 % of a wing's response to incidence: the loading dipped at the roots and the fits made it up outboard. It now carries the carry-through lift as wing. After the refit, ground effect with the propellers on came inside its limit (0.138 against 0.15) and flaps 20, unfitted, closed from 0.026 to 0.017 in C_m.
- The cruise propellers' swirl was spread evenly across the jet and turned only the strips' incidence, not their force. Against the power-on CFD, one propeller running as it does there, drag rose with thrust where the CFD's falls, and lift, pitch and the strips' own roll came to 0.73, 0.39 and 0.71 of the CFD's. Given the Goldstein loading NASA's CFD gives them, and the force turned with the swirl, they come to 1.07, 0.88 and 1.03, and drag to 1.01, nothing fitted (section 3.4). The climbs rose about 80 ft/min with both cruise motors running, 20 to 40 with one out or a bus lost: three DiTTo climbs that passed now miss, two that missed pass.

A comparison before that ran both propellers against the CFD's one, and read the lift as twice the CFD's.

- The ground-effect check and the fit read the centre of gravity's height before the model had first run, when it reads 0, so the first point of each set had its wing almost on the ground (0.006 of the span, not 0.181). That made two power-off misses at flaps 10 (0.060 and 0.123 against 0.05) and the high-lift one at 0.312, and put a wrong point in the high-lift fit's average. Read after the first run, power off matches and the high-lift miss is 0.177.
- The aileron's increments were read at each strip's own incidence and multiplied by 1.5 in lift and 2.0 in pitch. Where an up aileron's lifting line lifts its strips 7 to 8°, NeuralFoil's increment fades, and the gain made a deflected section's lift slope 0.18 a degree, so the strips took back more of it: the pitching moment stopped growing past −15°. Held at zero lift with no gain, roll and lift stay within their limits and the pitch error falls from 52 to 30 %.
- The tail's downwash came from two lumps of the wing's lift, inboard and outboard of the flap's end, so a right aileron's lost lift and the loss its lifting line spreads inboard all but cancelled at the tail: 0.007°, where summing each strip's vortices gives 0.19. Six lumps, three a side, give 0.21. Strip by strip, the blown flap's ends put single tail strips 25° up or down. The lumps first read the wake straight back along the body, so in a sideslip a blown wing's asymmetry fell on the wrong tail strips, 7° apart at 20°; carried with the air it is 2°. The P2006T's aileron per degree of sideslip, which that had brought inside its limit at 0.66, is back at 0.75.
- The rudder travelled 28°, the CFD's sweep. The P2006T's manual, NASA's ground simulation and its Mod III pilots' traces give 26.
- Every tyre had JSBSim's default side force, which makes the mains twice as stiff as measured tyres of their size, and the nose wheel was rigid on the pedals. With the pedals held straight, a 15 kt quartering tailwind rolled the aircraft onto a tip propeller from 40 kt.
- The power-on ground fit was averaged over the speeds. The model's own ground effect is right where the blowing is light, so the average took error off the slow cases and put it on the fast ones. Scaled by the discs' thrust coefficient it is nil unblown and grows with the blowing, as the shortfall does.
- Toward the stall the aileron's increment was handed back to NeuralFoil's own, whose flap at −25° climbs 0.13 a degree from 8 to 14°, twice the plain section's. The up aileron there kept 0.77 of the CFD's roll and the down one 1.5 times it. The section is now the plain one shifted by each deflection's equivalent angle, keeping the share of the increment at maximum lift that NACA measured on a plain flap [30]: the pitch error fell from 30 to 20 %, roll from 21 to 14, and at 8° and −25° roll went from 0.68 to 0.97 of the CFD's and pitch from 0.58 to 0.87. A share of 2/3, taken from the slotted flap's fit, passed too, but it was borrowed, and it left the down aileron near stall 1.75 times the CFD's.
- The rates' correction was taken only at the table's angles, which have none between 14 and 18°, while the strips stall one at a time within a degree, so near the stall the roll damping jumped between those points: −0.63 at 15°, flaps 10, against the tunnel's −0.28, and +0.09, none at all, at 13°, flaps 30. Taken on the tunnel's own curve, digitised a degree apart, and from 10° every quarter degree, it follows the tunnel from 12 to 18° within 0.01 to 0.02 rms at 100 kt and 0.01 to 0.04 at 60 kt, from 0.06 to 0.24. Larger roll rates still reach past single strips' stall, and past 18°, where the tunnel stops, the strips take the damping to nil at flaps 0 and 20°: that is the wing rock near the stall.
- The wing beam carried the nacelle as a point mass. Given the nacelle's rotary inertia, estimated from its size, second bending with the tip masses came within 8 % of the vibration test, from 17 % high. The clean wing's antisymmetric first bending moved from 5.0 to 5.2 % high and now misses its 5 % limit.

7 Assessment

38 checks miss. They fall into four kinds:

- **A source's figure in doubt.** The P2006T flight paper's short period and lift and moment slopes. That aircraft flies, so the reasons given can be weighed against other flight data.
- **Sources that can't all be met.** The high-lift slipstream, whose measured speed through the strip wing gives lift the CFD doesn't, so the model follows the lift; the climbs with both cruise motors running, 70 to 170 ft/min above the simulator's, whose cruise propellers carry an unpublished installed loss this model doesn't; the DiTTo motor-out climbs against the published motor tables, a mismatch inside the published figures that no flight can settle; the Mod II simulator's Dutch roll and phugoid against the flight test.
- **Limits written for another model.** Several mode limits written for the table model, against a simulator whose propellers have no in-plane force and whose roll seems to lack the blown wing's damping.
- **Real or open.** The slipstreams' own ground effect, fitted on their thrust coefficient from one solver; a 12 % high lift slope on the P2006T; the aileron's roll near stall, 27 % against 25; power-on roll from sideslip, 26 % against 25; the nose wheel's steering at speed, which NASA's simulator has far softer than geometry and measured tyres allow; the clean wing's antisymmetric first bending, 5.2 % high against 5; $C_{Y\beta}$; the Mod II simulator's roll time constant.

The motors' and rotors' inertias, loads, the torque rate and the speed-loop gains are estimates. Not modelled: torsion and flutter, the slipstreams' width changing with thrust and their path over the wing with incidence, dynamic inflow, the motors' temperatures. Mods III and IV never flew, so a pass against CFD or a simulator shows agreement with a prediction and no more, and a miss doesn't show which of the two is wrong. This work isn't connected with or endorsed by NASA, Tecnam or the aircraft's makers.

8 Conclusions

- A built-up model of the X-57 Mods III and IV, each of its fourteen propellers a disc with its own speed, its wing strips in their own slipstreams and its wing bending, runs in plain JSBSim at 120 Hz, generated from published data.
- Proved on the P2006T it passes the flight manual, and a second source shows the flight test's short period to be a method artefact.
- A ring-vortex slipstream with one constant does as well as a jet with three, and with none it halves the error of a plain

jet.

- The published motor failure is reproduced in time history, once flown in the clean configuration its own traces show.
- Limits were set first and never moved; one target was corrected for the published CG. 215 checks pass and 38 miss, each with its cause or marked open.

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